

Hakob Petrosyan

+374 (99) 011-971
jacpetrosyan@gmail.com

ML/DL Engineer

Portfolio
github.com/hakob-petro
linkedin.com/in/jacpetro

ML engineer specializing in the LLM training and inference stack: distributed training, parameter-efficient fine-tuning, RL post-training, and inference optimization. MSc in Applied Mathematics and Computer Science from MIPT. Winner of Armenia's national olympiads in Mathematics, Physics, and Astronomy.

SKILLS

ML/LLM	Python, PyTorch, Jax, Flax, Optax, OpenCV, OpenGL, asyncio/threading, HF (Transformers, TRL, PEFT, etc.), MS-Swift, TorchTune, TensorRT, OpenAI API, PydanticAI, OpenAI Gymnasium, etc.
Systems	Git, $\LaTeX$, Linux, Docker, vLLM, TEI, Tailscale/Headscale, OpenMP, MPI, DeepSpeed/FSDP, Network Technologies [1–4 lvl. OSI], TPU/GPU fundamentals
Languages	English (upper-intermediate), Russian (fluent speaker), Armenian (native language)

TECHNICAL EXPERIENCE

RL Environments Engineer (remote)

Feb 2026 — June 2026

Stealth

San Francisco, CA, USA

- > Designed and shipped a strict quality-first RL training environments for MLE-domain agentic tasks (details under NDA), owning task design, grading rubrics, and verification logic end-to-end.
- > Mitigated critical failure modes—such as reward hacking, reward denial, and instruction ambiguity—while ensuring strict adherence to pedagogical objectives under optimization pressure.
- > Reviewed environments from other contributors.

Python Developer (remote)

July 2025 — Oct 2025

HighSky

London, UK

- > Worked on decentralized machine learning systems (Bittensor technology) competing in various incentive-driven challenges (subnets), achieving results that placed us at the top of the rankings. Details under NDA.

ML Engineer (on-site)

Jan 2025 — July 2025

Grid Dynamics (Apple contractor)

Yerevan, Armenia

- > Contributed as an Apple contractor to the development of a large-scale internal platform for distributed LLM training and inference. Delivered multiple proof-of-concepts later adopted by the core engineering team.

ML Engineer (remote)

Aug 2023 — July 2025

ScaleGenAI | TheAgenticAI

Palo Alto, CA, USA

- > Parallelized LLM training using Tensor, Pipeline, and Data parallelism techniques with PyTorch Pippy, Megatron-LM, DeepSpeed and FSDP.
- > Fine-tuned popular open-source large language models using parameter-efficient fine-tuning (PEFT) methods such as LoRA and qLoRA, across various training objectives including SFT, DPO, ORPO, and GRPO. Developed a custom chain-of-thought (CoT) generation pipeline, and trained domain-specific reasoning models for medical and chemical tasks using GRPO. Biggest clusters used: 5 nodes with 8xH100 and 3 nodes with 8xH200
- > Explored various speculative decoding techniques, including vanilla speculative decoding, Eagle 1/2/3, ReDrafter, and Medusa 2. Achieved a 1.7× to 3.5× improvement in throughput of the internal inference engine as a result.
- > Contributed to the development cycle of a cost-efficient, multi-cloud LLM training/inference backend. Developed cluster fetching/provisioning logic and a backend framework for LLM fine-tuning. Implemented minor features for specific cloud providers using their API (AWS, DataCrunch, ShadeForm, Runpod, VastAI).

Data Scientist (remote)

Feb 2022 — May 2023

Intelligent Solutions

Moscow(Skolkovo), Russia

- > Implemented color autocorrection of images using the X-Rite color palettes with OpenCV and Python colour module. Image white balance correction with U-Net.
- > Implemented an HTTP server, which was a linked server between the backend API and the TensorFlow-serving service.
- > Converted dry-surface images to wet-appearance renders using 3D modeling.
- > Classification of lithotypes using XGBoost and label propagation with self-learning.

ML Researcher Intern (on-site)

Oct 2020 — Jul 2022

Acronis

Moscow, Russia

- > Diagnosed root causes of common hardware failures of systems with Windows-based operating systems. Modeling the future behavior of selected characteristics of the system.

+374 (99) 011-971
jacpetrosyan@gmail.com

Hakob Petrosyan

ML/DL Engineer

Portfolio
github.com/hakob-petro
linkedin.com/in/jacpetro

OTHER EXPERIENCE

ML/DL Lecturer

Yerevan State University

Sep 2024 — Dec 2024

Yerevan, Armenia

> Voluntarily teaching ML/DL fundamentals to students in technical fields, aiming to advance AI education.

EDUCATION

Master, *Moscow Institute of Physics and Technology*

Aug 2022 — July 2024

- ◇ Faculty of Innovation and High Technology.
- ◇ Department of Digital Transformation Technologies.
- ◇ Research topic: "*RL based dynamic determination of stop-loss limits for orders*".

Bachelor, *Moscow Institute of Physics and Technology*

Aug 2018 — July 2022

- ◇ Faculty of Radio Engineering and Cybernetics.
- ◇ Department of Theoretical and Applied Informatics (Acronis).
- ◇ Research topic: "*Anomalies detection in the working of hardware components of PC*".

PROJECTS

Several pet-projects that were carried out independently for interest or at the university:

- > *A simple flappy bird game with NeuroEvolution of Augmenting Topologies.*
- > *Prediction of missing DNA nucleotides using bidirectional LSTM.*
- > *Writing custom PyTorch optimizer for projected gradient descent algorithm.*
- > *Implemented Hinton's forward-forward training algorithm in PyTorch.*
- > *A cache-friendly block matrix multiplication.*
- > *The numerical solution of the transport equation using MPI.*

CERTIFICATIONS AND COURSES

- Neural networks and computer vision (Samsung Research, Russia)
- CCNAv7: Introduction to Networks (Cisco)
- CCNAv7: Switching, Routing, and Wireless Essentials (Cisco)
- C++ Development Basics: White Belt (MIPT & Yandex)
- C++ Development Basics: Yellow Belt (MIPT & Yandex)
- C++ Development Basics: Red Belt (MIPT & Yandex)